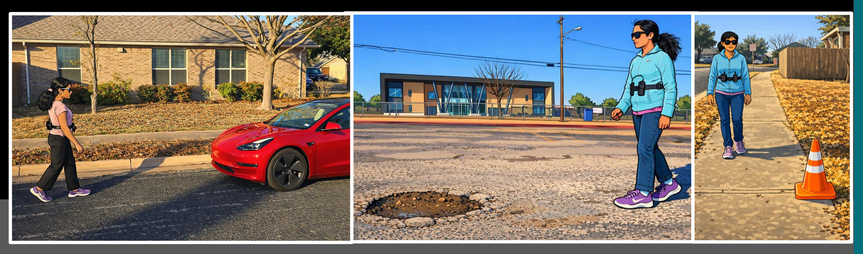
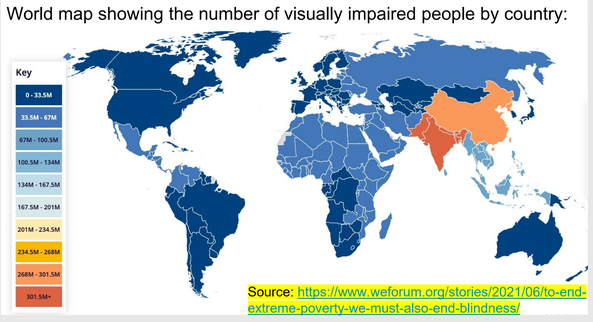
AI Navigo: Innovative Wearable for Visual Assistance



Project #: ETSD-0543-SR

Problem Statement

World map showing the number of visually impaired people by country:



Source: https://www.weforum.org/stories/2021/06/to-end-extreme-poverty-we-must-also-end-blindness/

Navigating daily life is a major challenge for visually impaired individuals. To empower visually impaired people by providing real time information about obstacles and the directions to navigate safely in a cost effective approach.

Research Question & Hypothesis

Research Question: Can machine learning be used to assist visually challenged people with navigation by detecting obstacles in their surroundings?

Hypothesis: Object detection frameworks, like YOLO models, can be used to develop a visual assistant device for detecting obstacles to aid visually challenged people with navigation

Background Research

Existing methods: cane, guide dog, or a sighted guide, AI powered solutions

Limitations:
- Guide dogs and sighted guides are costly,
- Cane has a limited detection range.
- Negatively impact an individual's sense of Independence.
- AI solutions are costly, complex and limited real time feedback

Research Gaps:
- Develop cost effective solution that provides real-time feedback and detects obstacles precisely

Dataset Description

Dataset Split:

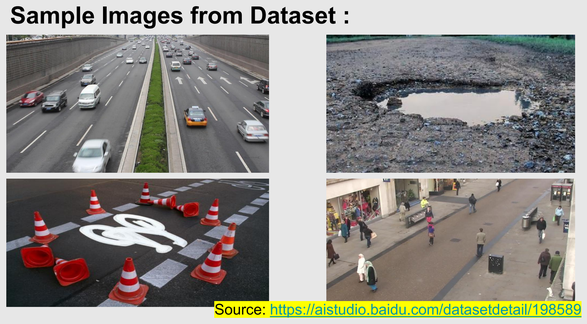
Train : 4393

Test : 628

Validation : 1255

4 classes : Traffic Cone, Pothole, people, Car

Sample Images from Dataset :



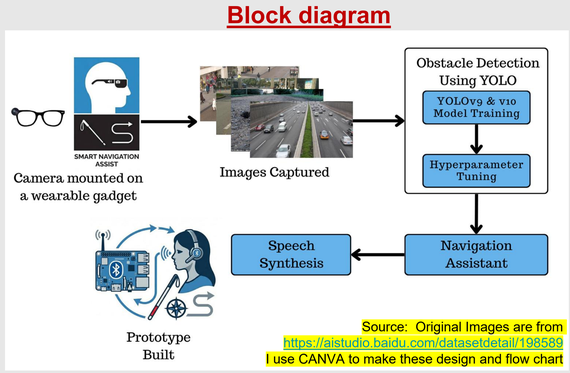
Source: https://aistudio.baidu.com/datasetdetail/198589

Materials

Components	Description
YOLOv9 and v10	Libraries required : Ultralytics
GPU	A100 Runtime
Platform	Google Colab Pro+
Storage	Google Drive
RaspberryPi 5	64-bit quad-core Arm Cortex-A76, 8GB board, 64GB Card
Endoscope Camera & glasses	Resolution: 1280x720 Frame Rate: 25fps Viewing Angle: 70 Degrees
Power bank	12000mAh 100W PD PPS
Bluetooth	Case Logic Wireless Earbuds

Procedure

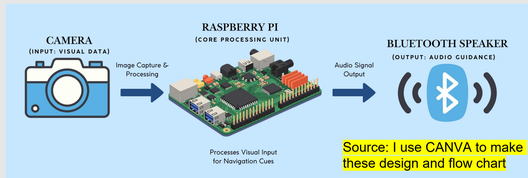
Block diagram



Source: Original Images are from https://aistudio.baidu.com/datasetdetail/198589
I use CANVA to make these design and flow chart

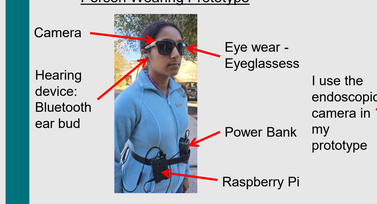
The sequential workflow of my project begins with the splitting of the dataset into train, test, and validation. The YOLOv9 and YOLOv10 models are trained on the training dataset to identify different features and patterns in the images. The hyperparameters of the model, such as Epochs and Learning rates, are fine tuned using the validation dataset, resulting in more optimized models. The models are evaluated on the test dataset using the performance metrics like precision, recall, F1-Score, mean Average Precision (mAP) to assess the functionality and accuracy of the model.

Prototype

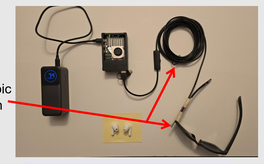


Source: I use CANVA to make these design and flow chart

Person Wearing Prototype



Prototype Image



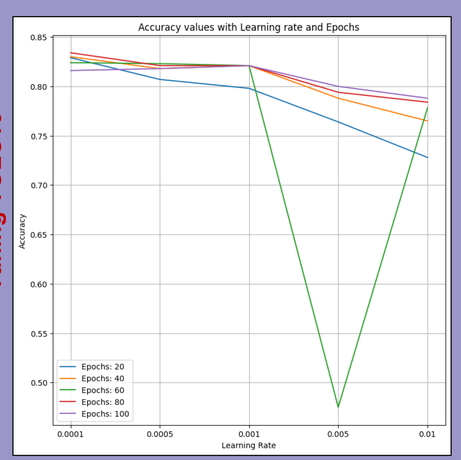
Source: I am the only person in the photo, and it was taken by my parents.

Metrics

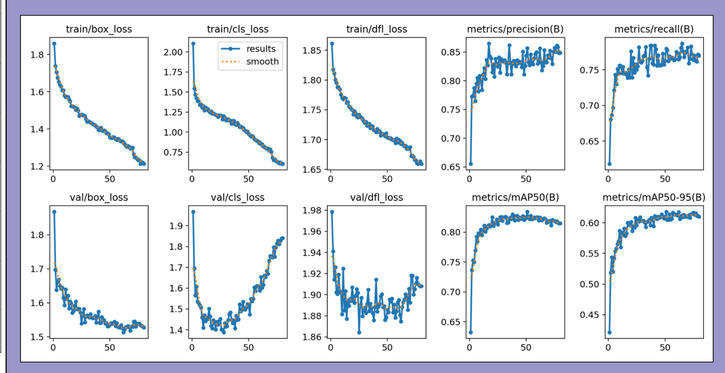
- **Box Loss :** Accuracy of predicted bounding boxes
- **Classification Loss :** Accuracy of object class predictions
- **Distribution Focal Loss :** Focuses on improving bounding box precision
- **IoU :** Overlap ratio between predicted and true bounding boxes
- **mAP50 :** Average Precision with 50 percentage IoU threshold
- **Precision :** correct positives out of predicted positives
- **Recall :** correct positives out of actual positives
- **F1 Score :** Harmonic mean of precision and Recall

Data/Observations

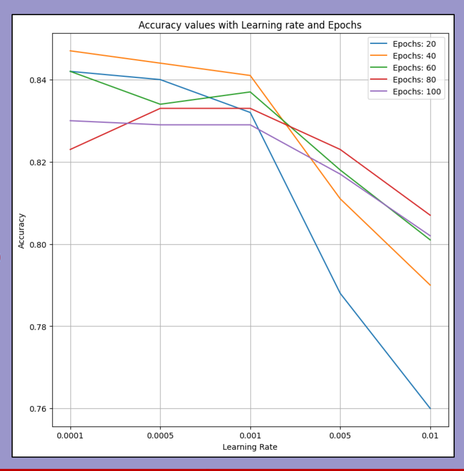
Hyperparameter Tuning YOLOv9



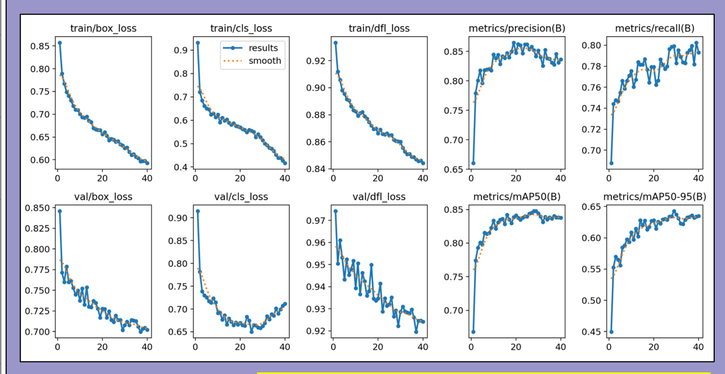
Loss Graph YOLOv9



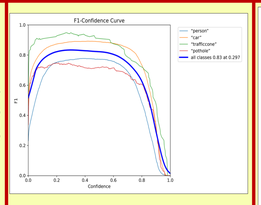
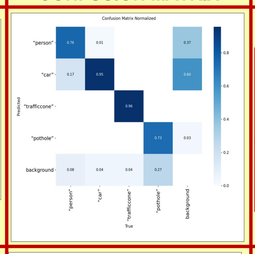
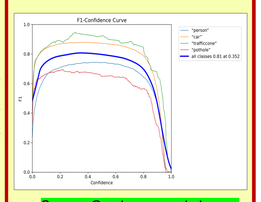
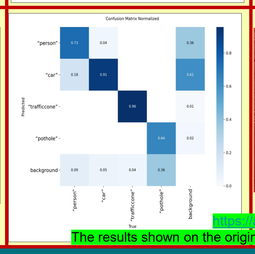
Hyperparameter Tuning YOLOv10



Loss Graph YOLOv10



Source: Graphs are made by me using my YOLO models

	F1-CURVE	CONFUSION MATRIX	TEST RESULT IMAGES
YOLOv9			
YOLOv10			

Source: Graphs are made by me using my YOLO models

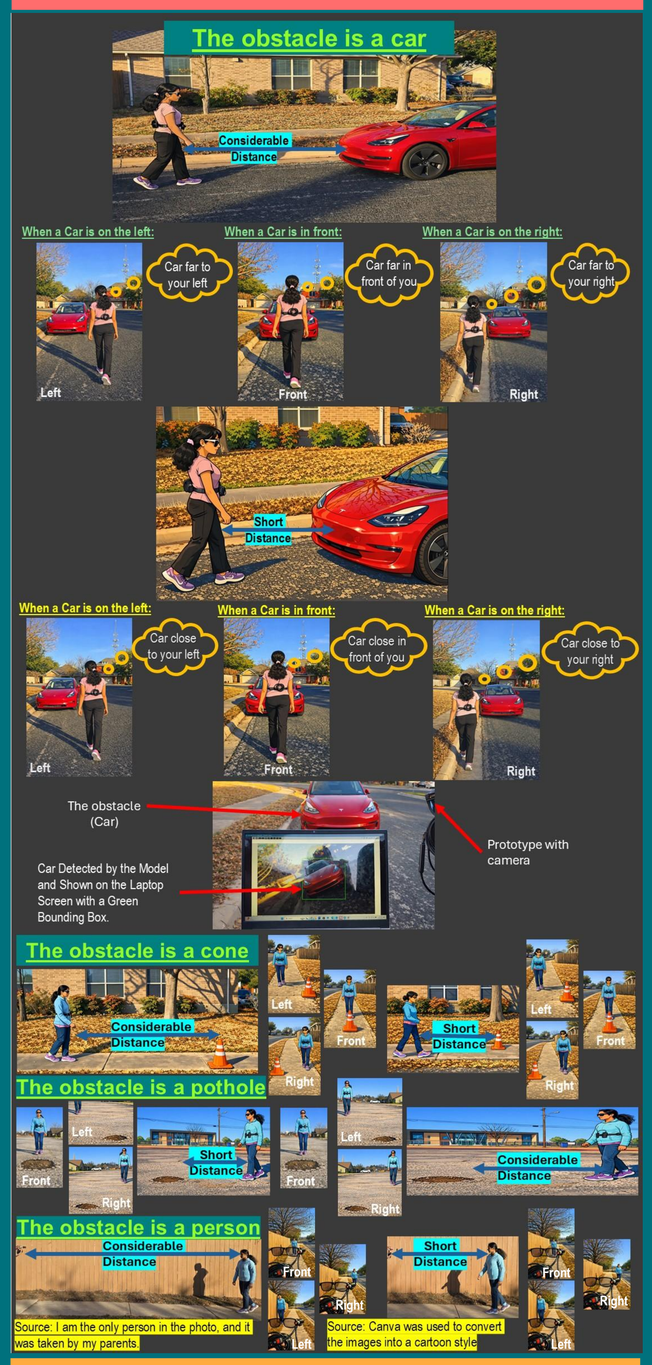
Source: Original Images are from https://aistudio.baidu.com/datasetdetail/198589

The results shown on the original photos were generated by my model.

Discussion

- For YOLOv9 : The model reached a peak F1 score of 0.83 for all classes with threshold of 0.297
- For YOLOv10 : he model reached a peak F1 score of 0.81 all classes with threshold of 0.352
- Both :
  - 30% of potholes were misclassified
  - 18% of persons instances were misclassified when located near cars
- Efficient Frames Per Second of 1.10

How Prototype Works



Conclusion

In conclusion, this project demonstrates that a YOLO model can be successfully integrated into a Raspberry Pi to detect obstacles and deliver audio feedback, enabling visually impaired individuals to navigate their daily environments more independently.

Created :
- YOLO model integrated into a Raspberry Pi with the ability to detect obstacles and provide a audio feedback

Challenges Addressed :
- Created a cost effective, real time model with a fps of 1.1
- Provides users more freedom of travel
- Solution with ability to detect obstacles from various distances

Limitations :
- Only four obstacles has been explored in this study
- Model makes errors in classification of obstacles or fails to detect them

Future Research

- Improve model's ability to detect current classes (potholes)
- Make prototype more compact
- Expand prototype to identify other obstructions
- Coordinate with relevant organizations to bring this project into real-world use

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Acknowledgement

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